Measures of Center

Students are introduced to mean, median and mode(s) and consider which of these measures of center best describes various quantitative data.

Lesson Goals	Students will be able to… • Students explore the concept of center of a distribution, learning how to compute the mean, median and mode(s) of a dataset • Students find the mean, median and mode(s) of various columns in the Animals table
Student-facing Lesson Goals	• Let's use mean, median, and mode to describe our data.
Prerequisites	• Simple Data Types • Contracts • Histograms • Visualizing the "Shape" of Data
Materials	• Lesson Slides • Animals Starter File • Summarizing Columns in the Animals Dataset (Page 23) • Mode(s) (Desmos) • *Optional: Data Cycle: Measures of Center* • *Optional: Critiquing Written Findings* • **Classroom visual:** Language Table
Preparation	• All students should log into code.pyret.org (CPO) and open their saved "Animals Starter File". If they don't have the file, they can open a new one from Animals Starter File.

Glossary

mean :: a representation of the center, or 'typical' value in a set of numbers, calculated as the sum of those numbers divided by the number of values.

median :: the middle element of a quantitative dataset

mode :: the most commonly appearing categorical or quantitative value or values in a dataset

outlier :: observations whose values are very different from the other observations in the same dataset, perhaps due to experimental error. Outliers can also be indicative of data belonging to a different population from the rest of the established samples.

quantitative data :: number values for which arithmetic makes sense

skew :: lack of balance in a dataset's shape, arising from more values that are unusually low or high. Such values tend to trail off, rather than be separated by a gap (as with outliers).

Mean *15 minutes*

Overview

Students learn about ***mean*** (or "average"), and how it is one way (among others!) to summarize a ***quantitative*** column.

Launch

According to the Animal Shelter Bureau, the average pet weighs almost 41 pounds.
Some medicines are dosed by weight: heavier animals need a larger dose that could be dangerous for smaller animals. If someone from the shelter needs to give a dose of medicine to the animals, is the "average" the best estimate we can use?
"The average pet weighs 41 pounds" is a statement about the entire dataset, which summarizes a whole column of values with a single number. Summarizing a big dataset means that some information gets lost, so it's important to pick an appropriate summary. Picking the wrong summary can have serious implications! Here are just a few examples of summary data being used for important things. Do you think these summaries are appropriate or not?

- Students are sometimes summarized by two numbers — their GPA and SAT scores — which can impact where they go to college or how much financial aid they get.
- Schools are sometimes summarized by a few numbers — student pass rates and attendance, for example — which can determine whether or not a school gets shut down.
- Adults are often summarized by a single number — like their credit score — which determines their ability to get a job or a home loan.
- When buying uniforms for a sports team, a coach might look for the most common size that the players wear.

Can you think of other examples where someone uses a number or two to summarize something complex?

Every kind of summary has situations in which it does a good job of reporting what's typical, and others where it doesn't really do justice to the data. In fact, the shape of the data can play a huge role in whether or not one kind of summary is appropriate!
One of the ways that Data Scientists summarize quantitative data is by talking about its *center* - literally asking "what is a typical value in this sample?", in the hopes of inferring something about a larger population. But there are many different ways to define "center", and each method has strengths and weaknesses. Let's check the "41 pounds" claim and see if it's an appropriate measure of center. Later on, you'll have a chance to apply what you've learned to your own dataset, to find the best way to provide an overall summary of the data.

Kinesthetic Activity

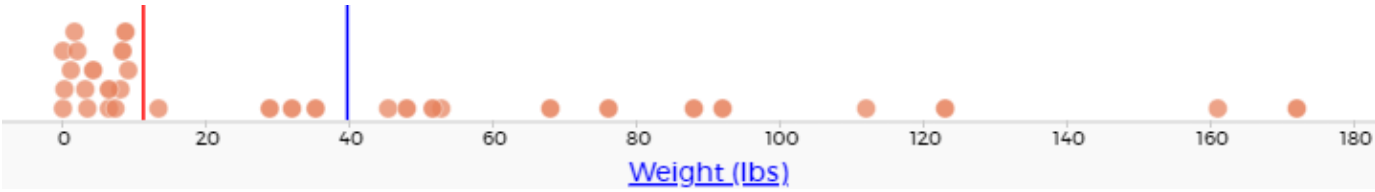
If you have a set of rulers, divide the class into groups such that every group has a ruler. Give each group 4-8 pennies and make sure every group has at least one pen or pencil.
The arithmetic mean is the number that "balances" all the other numbers in the sample. So let's do some real balancing!

1. The ruler represents a number line with values (weight) distributed equally across the line. If there's values at every inch from 0 to 12, where should the pencil be placed in order to balance the ruler on top of it?
2. Place a penny at 1 and 11. Where must the pencil be placed to balance those two values? What is the mean of of the values [1, 11]?
3. Place pennies at 1, 9 and 11. Where must the pencil be placed to balance those two values? What is the mean of of the values [1, 9, 11]?
4. Suppose you were to place two pennies at 2, and a third penny at 8. Can you *predict* where the pencil should be placed?

Investigate

Before digging into a discussion of mean, let's look at a visual.

If we plotted all the animals' weights as points on a number line, it would look something like this:



- Do you think there is a midpoint?
  - *There are 32 animals - meaning that there is not one point in the middle.*
- Is there a point that shows up most often?
  - *Since we see that dots are stacked up, it seems likely that there is some repetition in the animals' weights.*
- What do you think the red line represents? How about the blue one? How do you know?
  - *Be sure to solicit students' answers, which may vary. Red is the median, blue is the mean. Uncertainty at this point is okay! The remainder of the lesson is an exploration of these concepts.*

Each of these are different ways of "measuring center".

The Animal Shelter Bureau used one method of summary, called the ***mean***, or "average". The mean of a dataset is the sum of values divided by the number of values. To take the average of a column, we add all the numbers in that column and divide by the number of rows.

Pyret has a way for us to compute the mean of any ***quantitative*** column in a Table. It consumes a Table and the name of the column you want to measure, and produces the mean — or average — of the numbers in that column.

```
# mean :: Table, String -> Number
```

- What is the function's name? Domain? Range?
  - *The function's name is* `mean`*. The function consumes a table and string (domain), and produces a number (range).*

Notice that calculating the mean requires being able to add and divide, so the mean only makes sense for quantitative data. For example, the mean of a list of Presidents doesn't make sense. Same thing for a list of zip codes: even though we can divide a sum of zip codes, the output doesn't correspond to some "center" zip code.

- Type `mean(animals-table, "pounds")`. What does this give us?
  - *39.715625.*
- Does this support the Bureau's claims?
  - *No, the mean is less than 41 pounds.*
- Now, turn to Summarizing Columns in the Animals Dataset. In the "measures of center" section, fill in the computed mean.

Median *15 minutes*

Overview

Students learn a second measure of center: the ***median***. They learn the algorithm and the code to find the median, as well as situations where taking the median is more appropriate than the mean.

Launch

You computed the mean of that column to be almost exactly 41 pounds. That IS the average, but if we scan the dataset we'll quickly see that most of the animals weigh less than 41 pounds! In fact, more than half of the animals weigh less than just 15 pounds. What is throwing off the average so much?

Kujo and Mr. Peanutbutter!

In this case, the mean is being thrown off by a few extreme data points. These extreme points are called ***outliers***, because they fall far outside of the rest of the dataset. Calculating the mean is great when all the points are fairly balanced on either side of the middle, but it distorts things for datasets with extreme outliers. The mean may also be thrown off by the presence of ***skewness***: a lopsided shape due to values trailing off to the left or right.

- Make a `histogram` of the `pounds` column, and try different bin sizes.
- Can you see the huge number of animals clumped to the left, with Kujo and Mr. Peanutbutter as outliers skewed to the right?

A different way to measure center is to line up all of the data points — in order — and find a point in the center where half of the values are smaller and the other half are larger. This is the ***median***, or "middle" value of a list.

As an example, consider this list of ACT scores:

`25, 26, 28, 28, 28, 29, 29, 30, 30, 31, 32`

Here 29 is the ***median***, because it separates the "bottom half" (5 values below it) from the top half" (5 values above it).

The algorithm for finding the median of a quantitative column is:

1. Sort the numbers
2. Cross out the highest and lowest number
3. Repeat until there is only one number left…
4. When there are an even number of numbers in the list, as in the example below, there will be two numbers left at the end. Take the *mean* of those two numbers.

`3, 7, 9, 21`

The median of this list is 8, because 8 is the mean of the two middle numbers, 7 and 9. To find their mean, we added 7 and 9 to get 16 and split 16 in half.

Investigate

- Pyret has a function to compute the median of a list as well. Find the contract in your contracts page.
- Compute the median for the `pounds` column in the Animals Dataset, and add this to [Summarizing Columns in the Animals Dataset](#).
  - *The median is 11.3.*
- Is it different than the mean?
  - *Yes, it is very different!*

- What can we conclude when the mean is so much greater than the median?
  - *There are some very heavy animals that are causing the mean to be higher.*
- For practice, compute the mean and median for the weeks and age columns.
  - *Weeks: mean - 5.75; median - 4. Age: mean - 4.359375; median - 3.*

Synthesize

By looking at the histogram, we can see that it's probably better to use the mean or median.
Strong left skewness and/or low outliers can pull the mean down below the median, while right skewness and/or high outliers can pull it up above the median.
Mean is generally the best measure of center, because it includes information from every single point. But it's inaccurate for highly-skewed datasets, so statisticians fall back to the median.

Modes *25 minutes*

Overview

Students learn about the mode(s) of a dataset, how to compute the mode, and when it is appropriate to use this as a measure of center.

Launch

The third measure of center is called the ***modes*** of a dataset. The ***modes*** of a dataset are the values that appear *most often*.
Median and Mean always produce one number and many datasets are what we call "unimodal", having just one mode. But sometimes there are exceptions!

- If two or more values are equally common, there can be more than one mode.
- If all values are equally common, then there is no mode at all!

Consider the following three datasets:

```
1, 2, 3, 4
1, 2, 2, 3, 4
1, 1, 2, 3, 4, 4
```

- The first dataset has *no mode at all!*
- The mode of the second dataset is 2, since 2 appears more than any other number.
- The modes (plural!) of the last dataset are 1 and 4, because 1 and 4 both appear more often than any other element, and because they appear equally often.

Mode is rarely used to summarize quantitative data. It is very common as a summary of *categorical* data, telling us which category occurs most often.
In Pyret, the mode(s) are calculated by the modes function, which consumes a Table and the name of the column you want to measure, and produces a *List* of Numbers.

```
# modes :: Table, String -> List<Number>
```

Investigate

- Compute the `modes` of the `pounds` column, and add it to [Summarizing Columns in the Animals Dataset](). What did you get?
  - *0.1 and 6.5*

Synthesize

The most common animal weights are 0.1 and 6.5! That's well below our mean and even our median, which is further evidence of outliers or skewness.
At this point, we have a lot of evidence that suggests the Bureau's use of "mean" to summarize animal weights isn't ideal. We have three reasons to suspect that ***mean*** isn't the best value to use:

- The median is only 11.3 pounds.
- The modes of our dataset are 6.5 pounds and 0.1 pounds, which suggests a cluster of animals that weigh less than one-sixth the mean.
- When viewed as a histogram, we can see the right skewness and high outliers in the dataset. Mean is sensitive to datasets with skewness and/or outliers.

"In 2003, the average American family earned $43,000 a year — well above the poverty line! Therefore very few Americans were living in poverty."

Do you trust this statement? Why or why not?

- *Sample response: The mean is sensitive to outliers, and billionaires like Elon Musk, Jeff Bezos, etc. pull the mean is heavily to the right. This makes it appear that the "average" American family earns far more than they actually do. That's why the conclusion "very few Americans were living in poverty" cannot be drawn based on the mean.*

Consider how many policies or laws are informed by statistics like this! Knowing about measures of center helps us see through misleading statements.

You now have three different ways to measure center in a dataset. But how do you know which one to use? Depending on the shape of the dataset, a measure could be really useful or totally misleading! Here are some guidelines for when to use one measurement over the other:

- If the data doesn't show much skewness or have outliers, ***mean*** is the best summary because it incorporates information from every value.
- If the data has noticeable outliers or skewness, ***median*** gives a better summary of center than the mean.
- If there are very few possible values, such as AP Scores (1–5), the ***mode*** could be a useful way to summarize the dataset.

NOTE: We strongly recommend having students practice the Data Cycle with measures of center, using Data Cycle: Measures of Center. Sometimes what's created isn't a table *or* a display, and this activity demonstrates that. It also drives home an important difference between Arithmetic and Statistical Questions.

Additional Exercises

- Mode(s) (Desmos)
- Critiquing Written Findings
- Data Cycle: Measures of Center